



Software Engineering Department

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Capstone Project – Phase A

Epileptic Seizure Detection and Prediction using EEG

Team Code: 26-2-R-7

GitHub:

<https://github.com/IbrahimAsad421/EEG-Based-Seizure-Prediction>

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Abstract

Epileptic seizures affect approximately 50 million people worldwide, with roughly one third of patients remaining medication-resistant. For these individuals, the unpredictability of seizure onset is a primary disability, driving the need for real-time prediction systems. While deep learning models offer high sensitivity, their computational demands and "black-box" nature hinder deployment on lightweight clinical wearables.

This project will build on the Complexity Collapse hypothesis: that a seizure is not the brain becoming louder but the brain becoming simpler — a collapse from high-entropy neural diversity into a single, rigidly synchronized mode. The hypothesis describes seizures rather than predicting them; it was established offline, on already-labelled recordings, and names real-time prediction as future work.

Our contribution will be Ψ , a deterministic collapse operator light enough to run on a wearable. Ψ scores each window of EEG along three axes — informational complexity, inter-regional coherence, and network topology — computed by streaming algorithms rather than the offline mathematics of the original. The weighted three-axis form follows an existing reference implementation; the axes and their computation are our own.

To compensate for the precision-efficiency trade-off, we will implement a Soft Fusion post-processing stage using a rolling temporal buffer and Top-K averaging to stabilize alarms and suppress artifacts. Because our axes are computed differently from those the reference implementation fused, we will compare its weights against a re-tuned variant learned by a transparent, shallow classifier (Logistic Regression, Linear SVM, or a small Decision Tree) under strict leave-one-subject-out cross-validation, and deploy whichever performs better.

Ψ will be evaluated on the CHB-MIT Scalp EEG Database with clinically relevant metrics — event-based F1, false predictions per hour, and Seizure Prediction Horizon. The result will be an interpretable, low-power operator that carries the Complexity Collapse principle from offline description to real-time wearable prediction.

Keywords: Electroencephalography (EEG), Seizure Prediction, Complexity Collapse, Hardware-Efficiency, Lempel–Ziv Complexity, Phase-Locking Value, Persistent Homology, Soft Fusion, Wearable Device.

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1. Introduction

For a person living with drug-resistant epilepsy, the most disabling feature of the illness is rarely the seizure itself — it is the impossibility of knowing when the next one will arrive. A system that could give even a few minutes of warning before seizure onset would return a measure of autonomy to a population for whom medication has already failed. The engineering problem this raises is unusual: such a system must be accurate enough to be clinically useful, interpretable enough to be clinically trusted, and light enough to run continuously on a device the patient is willing to wear — three constraints that the field has so far satisfied only in pairs, never all three at once. This chapter situates that problem, the need, existing solutions and our proposal .

1.1 The Clinical Need

Epilepsy is one of the most common serious neurological disorders in the world, affecting more than 50 million people of all ages [13]. Roughly one third of these patients suffer from drug-resistant epilepsy — a clinical category defined by the International League Against Epilepsy as the failure of two adequately chosen and tolerated anti-seizure medications to achieve sustained seizure freedom [7]. For this group, no pharmacological treatment reliably prevents seizures from occurring, and surgical intervention is feasible only for a minority of cases in whom a single, well-localized epileptogenic focus can be identified.

Unpredictability is what converts a chronic illness into a daily disability. Because seizures may strike without warning, drug-resistant patients are prevented from driving, swimming, working at heights, or being left alone with infants, and many describe the resulting loss of autonomy as more burdensome than the seizures themselves. A system that could detect the onset of a seizure minutes before it occurs — even with modest accuracy — would restore meaningful independence to this group.

The clinical requirements such a system must meet are demanding. It must run continuously on a wearable device the patient is willing to wear for months or years, which immediately rules out architectures that depend on GPU-class compute or cloud connectivity. It must produce alarms with very few false positives, because alarm fatigue rapidly causes patients and caregivers to ignore the device. It must give enough lead time before seizure onset for the patient to take a protective action — a Seizure Prediction Horizon of at least 1–5 minutes. And its decisions must be interpretable, so that neurologists can trust the system, regulators can approve it, and engineers can diagnose its failures.

1.2 Existing Solutions

Existing approaches to seizure prediction can be grouped into three broad directions, each addressing part of the problem from a different angle:

- **Deep-learning predictors:** Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), the two dominant deep-learning families for time-series data, are trained directly on raw or lightly preprocessed EEG and report strong sensitivity on benchmarks such as CHB-MIT [11]. These models extract their own features automatically from large quantities of labelled data, which is both their strength and their weakness.
- **Spectral and statistical feature engineering:** A second family of methods relies on hand-crafted features computed from short EEG windows — for instance, the relative power of standard frequency bands (delta, theta, alpha, beta, gamma), the entropy of the power spectrum, or simple measures of phase alignment between channels. These features are physiologically motivated, computationally cheap, and can be implemented on modest hardware. Surveys of CHB-MIT report a long line of such systems [11].
- **Theoretical complexity frameworks:** A third, more recent direction approaches the problem from the side of theoretical neuroscience rather than pattern recognition. The Complexity Collapse hypothesis, formalized in 2025 [9], argues that seizure onset is not the brain becoming louder but the brain becoming simpler: a catastrophic shift from the high-entropy, weakly-correlated activity of the healthy state into a single rigidly synchronized mode. The hypothesis identifies three measurable facets of this collapse — a decrease in signal complexity, a change in inter-regional connectivity, and a flattening of the underlying network topology. The hypothesis itself defines no operator; a separate reference implementation [6] fuses three such axes into a single weighted score, the operator form this project builds on (Section 1.4).

1.3 The Problem with Existing Solutions

Each of the three directions above addresses part of the seizure-prediction problem but leaves the others unsolved. Taken together, three persistent gaps obstruct the deployment of seizure prediction for the patients who need it most.

- **The interpretability gap:** Deep-learning predictors operate as opaque function approximators. A neurologist cannot inspect why a given alarm was raised, which limits clinical trust, slows regulatory approval, and prevents the feature-level error analysis that engineers need to improve the system over time. Spectral and statistical methods are more transparent, but they typically operate on a single mathematical axis (frequency, coherence, or complexity) and so reveal only one facet of pre-ictal change.

- **The hardware gap:** State-of-the-art deep architectures depend on GPU-class compute for both training and inference. Wearable monitoring devices — the only deployment target that makes prediction useful outside the hospital — operate under strict constraints on power, memory, and real-time latency that these architectures cannot meet. Even some hand-crafted feature pipelines, while cheaper, still rely on floating-point operations and full-precision spectral transforms that are awkward on embedded hardware.
- **The translation gap:** Theoretical frameworks such as the Complexity Collapse hypothesis [9] offer principled, interpretable, multi-axis descriptions of what is going wrong in the brain before a seizure. However, they are formulated for offline analysis on complete recordings: brain states are characterized over a full-length recording and compared after the fact with statistical tests. These operations assume full-precision arithmetic and unrestricted access to the entire recording, neither of which holds on a streaming wearable. A theory that cannot run on the device a patient actually wears does not, for that patient, prevent any seizures.

No single existing approach closes all three gaps at once. Deep learning sacrifices interpretability and hardware feasibility for sensitivity; spectral features sacrifice multi-axis coverage for simplicity; the Complexity Collapse framework offers principled multi-axis coverage but only in an offline, full-precision form. The opportunity this project pursues is to take the most theoretically principled of the three directions and re-engineer it so that it also satisfies the interpretability and hardware constraints the other two leave unmet.

1.4 Our Proposed Solution

We propose Ψ , a deterministic, hardware-efficient collapse operator that turns the Complexity Collapse principle into a real-time predictor, evaluated on the CHB-MIT Scalp EEG Database. Ψ scores each window of EEG along three axes — informational complexity, inter-regional coherence, and topological structure — each computed by an algorithm chosen to stream on embedded hardware, and fuses them with weights learned from labelled data. Ψ adopts the collapse principle of [9] and the weighted three-axis operator form of Kadhim's $\Delta\Phi$ [6], but reproduces neither: [9] defines no operator and evaluates its axes offline, while [6]'s operator computes its axes from raw-amplitude proxies.

The pipeline keeps the three-axis structure — informational complexity, inter-regional coherence, and topological structure — but selects a deterministic, streaming-friendly

algorithm for each axis rather than inheriting the offline methods of [9] or the amplitude proxies of [6]:

- The **informational axis** is computed as Multiscale Lempel–Ziv Complexity [5,14] over binary streams produced by SURE thresholding [2,12] and Hilbert-phase extraction [4]. This is our choice of informational measure; the original hypothesis uses permutation entropy [9], which we do not adopt.
- The **coherence axis** is computed as the Phase-Locking Value [8] from the phase-binarized streams. PLV is an addition of ours: the original characterizes connectivity through Pearson correlation [9], not phase locking, and we introduce PLV as an explicit, streaming-friendly coherence axis.
- The **topological axis** is computed from a similarity matrix of lightweight per-window channel summaries, reduced to three scalar summaries of the persistence diagram (total persistence, maximum persistence, and bar count). The original characterizes network structure with both persistent homology and a discrete Ricci curvature [9]; we do not adopt the Ricci-curvature geometry, which does not stream — our topological axis rests on persistent homology alone. The scalar reduction is our hardware-efficiency choice; the original retains the full diagram.

We introduce a transparent machine-learning layer (Logistic Regression, Linear SVM, or a small Decision Tree) that re-tunes these weights from labelled CHB-MIT data, and Sections 6 and 7 compare the re-tuned operator against the weights of [6] under strict leave-one-subject-out cross-validation. To suppress isolated noise spikes and stabilize alarms, the operator score is passed through a Soft Fusion post-processing layer based on a rolling Top-K average [1] before any alarm decision is made. Critically, this re-tuning is a one-time, offline training step. At deployment, the operator reduces to three multiplications and two additions per window — no model inference, no gradient computation, no GPU, the per-axis cost reduction that produces the three precomputed features is quantified in Section 3.4. The interpretability and hardware critiques of Section 1.3 concern deep architectures performing inference at every window, not a fixed linear combination of three precomputed features.

The intended contribution is not to outperform deep-learning predictors on raw sensitivity, but to demonstrate that an interpretable, low-power, theoretically grounded operator can reach clinically useful performance — measured by Event-Based F1, False Prediction Rate per hour (FPR/h), and Seizure Prediction Horizon (SPH/SOP) [1,11] — under the resource constraints of real wearable hardware. Concretely, this project produces three deliverables:

- (i) a realization of the $\Delta\Phi$ operator form on binarized streams.
- (ii) a re-tuned, cross-validated variant of the operator with learned weights.
- (iii) a head-to-head comparison between the two on the full 22-patient CHB-MIT cohort, reported with clinically relevant metrics.

2. The Dataset

2.1 Source and Rationale

This project uses the CHB-MIT Scalp EEG Database, a publicly available collection of long-term scalp EEG recordings released by Boston Children’s Hospital and MIT and hosted on PhysioNet [3]. The dataset has become a standard benchmark in the seizure detection and prediction literature [11]. We adopt it for two reasons: it is the dataset the original Complexity Collapse framework [9] drew its records from, and its multi-day continuous recordings from drug-resistant pediatric patients realistically expose the long inter-ictal stretches that any wearable predictor must cope with in deployment.

2.2 Cohort and Recording Specifications

The database contains 664 EDF files organized into 23 cases from 22 unique subjects: 5 males (ages 3–22) and 17 females (ages 1.5–19) [3,10]. One female subject appears in two cases, chb01 and chb21, recorded approximately 1.5 years apart [3,10] — a distinction that matters for cross-validation and is addressed in Section 2.4. All recordings were collected during pre-surgical evaluation, while patients were monitored for several days following the withdrawal of anti-seizure medication [3]. The principal recording parameters are given in Table 2.1.

Table 2.1 — CHB-MIT recording specifications [3].

Parameter	Value
Sampling rate	256 Hz
Resolution	16-bit
Electrode placement	International 10–20 system
Montage	Bipolar (double-banana)
Channels per file	Predominantly 23; a few 24–26
File duration	Predominantly 1 hour; some 2–4 hours
File format	European Data Format (EDF)

2.3 Seizure Annotations

The dataset contains 198 annotated seizures across the 23 cases [3]. Each seizure was reviewed by clinical neurologists at Boston Children’s Hospital, who marked its onset and termination timestamps to the nearest second [3,10]. Of the 664 files, 129 contain one or more seizures; the rest are entirely inter-ictal [3]. These precise annotations are what make the supervised training stage of our pipeline (Section 6) possible: from the onset timestamp we derive the pre-

ictal window — the prediction target of this project — against which the classifier is trained and against which Event-Based F1, SPH, and SOP are computed.

2.4 Characteristics Affecting Pipeline Design

Four properties of CHB-MIT directly constrain pipeline design and must be acknowledged before windowing and preprocessing are specified.

- **Non-uniform channel sets:** Some files contain non-EEG channels (ECG, VNS trigger) that would corrupt every feature axis if treated as EEG; others contain 24–26 channels rather than the usual 23 [3]. Preprocessing therefore excludes non-EEG labels and restricts each subject to a common channel subset so that the per-window feature vector has consistent dimensionality.
- **One subject in two cases:** chb01 and chb21 are the same subject (Section 2.2). To avoid patient leakage in leave-one-subject-out validation, we merge them into a single cross-validation fold: whenever this subject is held out, both cases are held out together. The two cases remain separate recordings during feature computation; only the fold grouping is affected.
- **Inter-file gaps are not always short:** PhysioNet documents that gaps between consecutive EDF files are usually ≤ 10 seconds but can be substantially longer [3]. A sliding window must not straddle a file boundary, and pre-ictal labels must not be assigned across an unknown gap; each file is therefore windowed independently, with the gap treated as missing data.
- **Severe class imbalance:** Of the 664 EDF files, only 129 contain any seizures, and within those files the ictal periods themselves span less than 1% of total recording time [3,11]. Pre-ictal periods add a further few percent, depending on the chosen pre-ictal window length; the remainder is inter-ictal. Per-window accuracy is therefore meaningless as an evaluation metric, which is why our verification plan uses event-based metrics.

2.5 Access and License

The dataset is freely available at <https://physionet.org/content/chbmit/1.0.0/> under the Open Data Commons Attribution License v1.0; all protected health information has been de-identified [3]. Use requires citation of both the original source [10] and the PhysioNet resource paper [3].

3. Definitions

This section establishes the terminology used throughout the rest of the book. Figure 3.1 below gives a high-level view of the pipeline as a signal-flow diagram, showing how each window of EEG is transformed into an alarm decision; the rest of this chapter then defines each block in turn. The chapter is organized into four groups: the clinical and theoretical

foundations of the project (3.1), the mathematical objects used in the prediction pipeline, presented in the order in which they are applied to the data (3.2), the methodology used to evaluate the resulting system (3.3), and the computational cost of the resulting pipeline (3.4). A reference table of all symbols introduced is given in Table 3.1.

Figure 3.1 — Signal flow through the proposed pipeline for a single window of EEG. Each block is defined in 3.2.

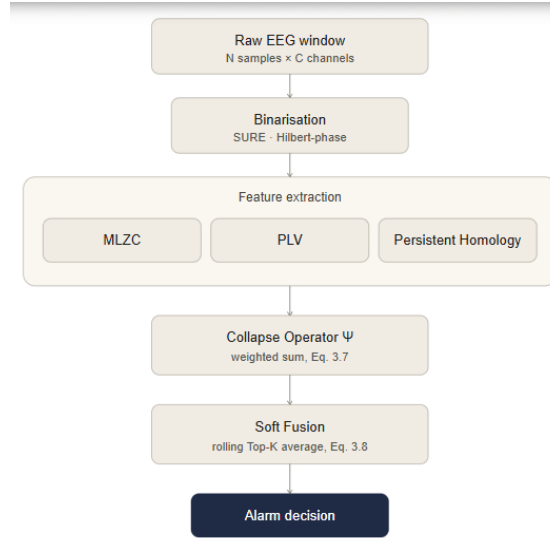


Table 3.1 — Notation used in this work.

Symbol	Name	Meaning / use
$x[n]$	EEG sample	Sampled scalp EEG value at sample index n (256 Hz).
$b[n]$	Binarized stream	Result of SURE or Hilbert-phase binarization of $x[n]$; $b[n] \in \{0,1\}$.
τ	Scale factor	Coarse-graining factor in MLZC ($\tau = 1, 2, \dots$).
$\phi[n]$	Instantaneous phase	Hilbert-derived phase of $x[n]$ at sample n .
$c(n)$	LZ token count	Number of distinct substrings in the first n samples of b .
N	Window length	Number of samples per analysis window (1,280 at 5 s).
C	Channel count	Number of EEG channels retained after preprocessing.
$P(t)$	Total persistence	Sum of (death – birth) over the persistence diagram.
$\Psi(t)$	Collapse Operator	Unified diagnostic score at time t (Eq. 3.7b).
μ, σ	Baseline mean and s.d	Per-patient inter-ictal baseline mean and standard deviation of each axis, estimated from training-fold windows only.
w_1, w_2, w_3	Operator weights	Positive coefficients combining the normalized axes (Eq. 3.7b)..
$S(t)$	Soft-Fusion score	Top-K average of recent Ψ values (Eq. 3.8).
W	Buffer length	Number of recent windows in the Soft-Fusion buffer.
K	Top-K parameter	Number of highest Ψ values averaged within W .
τ_{alarm}	Alarm threshold	Threshold on $S(t)$ above which an alarm is raised.
T_{pre}	Pre-ictal horizon	10 min = 153,600 samples at 256 Hz.

3.1 Clinical and Theoretical Foundations

Epilepsy and epileptic seizure: Epilepsy is a chronic neurological condition characterized by a persistent tendency to generate unprovoked, recurrent seizures — transient episodes of abnormal, excessive, and hypersynchronous neural activity [13]. An epileptic seizure is a single such event occurring in the context of this condition, distinguished from a provoked seizure (e.g. one triggered by acute fever or metabolic disturbance). All seizures in the CHB-MIT database used here are epileptic seizures from diagnosed pediatric patients.

Drug-resistant epilepsy: Also called refractory or intractable epilepsy; defined, with its ILAE criterion and prevalence, in Section 1.1 [7]. The patients in CHB-MIT belong to this category: recordings were collected during pre-surgical evaluation after medication had been withdrawn to provoke seizures for localization.

Ictal, pre-ictal, and inter-ictal states: Three clinically distinct phases of brain activity. The ictal state is the seizure itself. The inter-ictal state is the baseline period between seizures, representing normal or non-seizure brain activity well outside of the seizure event and its immediate transitional periods. The pre-ictal state is the window immediately preceding seizure onset; in this project we commit to $T_{\text{pre}} = 10$ minutes (justified in Section 7.1).

Complexity Collapse Hypothesis: A theoretical framework formalized in 2025 [9] that characterizes the transition from inter-ictal to ictal activity as a catastrophic loss of complexity: the healthy brain operates in a high-entropy regime with diverse, weakly correlated rhythms, while the seizure state corresponds to a collapse into a single dominant, rigidly synchronized mode. The original work quantifies this collapse offline through three descriptive facets — signal complexity, network connectivity, and network structure — compared with statistical tests; it defines no operator. Our project adopts these facets as the axes of its own Collapse Operator (Ψ , Section 3.2), instantiating them as MLZC, PLV, and persistent-homology summaries.

Electroencephalography (EEG): A non-invasive recording of the brain’s electrical activity, captured as a multi-channel time series sampled simultaneously across scalp electrodes. For CHB-MIT, approximately 23 channels are recorded at 256 Hz (Section 2.2).

3.2 Mathematical Pipeline

The pipeline is presented in the order in which operations are applied to the EEG signal: first binarization, then the three feature axes (MLZC, PLV, persistent homology), then their integration into the Collapse Operator, and finally Soft Fusion at the alarm-decision stage.

Binarization: The process of converting a continuous EEG sample sequence $x[n]$ into a binary string $b[n] \in \{0,1\}$. We use two binarization methods:

- SURE thresholding [2,12] selects an amplitude threshold τ_{SURE} that minimizes Stein’s Unbiased Risk Estimate over the window, then

$$b[n] = 1 \text{ if } |x[n]| > \tau_{\text{SURE}}, \text{ else } b[n] = 0$$

(3.1) SURE binarization.

We repurposed SURE as an automatic, adaptive threshold to separate signal spikes from the baseline. While not perfectly optimal, this per-window adjustment standardizes the data, allowing us to fairly compare complexity across different patients and timeframes.

- Hilbert-phase binarization [4] computes the analytic signal via the Hilbert transform, extracts the instantaneous phase $\phi[n]$, and outputs 1 when the phase is in the upper half of its cycle:

$$\phi[n] = \arg(x[n] + i\mathcal{H}\{x[n]\}); \quad b[n] = 1 \text{ if } \phi[n] \geq 0, \text{ else } 0$$

(3.2) Hilbert-phase binarization.

Phase-based binarization captures temporal synchronization rather than amplitude, which is the property most relevant to seizure dynamics — pre-ictal collapse manifests primarily as neurons locking into a shared firing rhythm [4].

Lempel–Ziv complexity (LZ76): A measure of the algorithmic complexity of a binary string, computed by parsing the string left to right and counting the number of distinct substrings $c(n)$ that have not been seen before. The normalized complexity is:

$$LZC = \frac{c(n) \log_2 n}{n}$$

(3.3) Normalized LZ76 complexity.

LZC approaches 1 for random sequences and tends toward 0 for repetitive ones; a pre-ictal complexity collapse manifests as a downward shift in LZC [4].

Multiscale Lempel–Ziv complexity (MLZC): An extension of LZC that computes complexity at multiple temporal resolutions [5]. Standard MLZC coarse-grains the continuous signal and then binarizes; our adaptation instead coarse-grains the already-binarized SURE stream $b[n]$, so the informational axis never needs the continuous EEG to be retained (consistent with the memory budget of Section 3.4). For scale factor $\tau \in \{1, 2, \dots, T\}$, the binarized stream is coarse-grained by summing non-overlapping windows of length τ :

$$y_\tau[k] = \sum_{i=(k-1)\tau+1}^{k\tau} b[i], \quad y_\tau[k] \in \{0, 1, \dots, \tau\}$$

(3.4) Coarse-graining of the binarized stream at scale τ .

LZC is then computed directly on the integer-valued sequence y_τ , so the LZ76 parse runs on integer sequences — additions and comparisons only, no floating-point multiply — at every scale ($\tau = 1$ recovers single-scale LZC on $b[n]$). MLZC exposes complexity collapses that occur preferentially in fast-frequency bands and would be masked in single-scale LZC by dominant slow waves [5,14]; prior work shows the pre-ictal complexity collapse manifests preferentially in the fine-scale frequencies that single-scale analysis cannot resolve [14], which is why MLZC replaces single-scale LZC as the informational axis of the Collapse Operator. Because this reorders binarization and coarse-graining relative to

standard MLZC [5], the effect of the reordering on multiscale sensitivity is a Phase B validation item.

Phase-Locking Value (PLV): A measure of the phase synchronization between two EEG channels a and b over a window of N samples, computed from their instantaneous phases $\phi_a[n]$ and $\phi_b[n]$ [8]:

$$PLV_{a,b} = \left| \frac{1}{N} \sum_{n=1}^N e^{i(\phi_a[n] - \phi_b[n])} \right|$$

(3.5) Phase-Locking Value.

PLV ranges from 0 (completely independent phases) to 1 (perfectly locked phases). Pre-ictal hyper synchronization manifests as elevated PLV between distant electrode pairs [8]. In our adaptation PLV will be computed between phase-binarized streams [4] rather than between continuous phase angles, reducing the per-sample inner loop from complex multiplication to bitwise comparison.

Persistent homology: A tool from Topological Data Analysis that tracks the appearance and disappearance of topological features — connected components (H_0) and loops (H_1) — across a sequence of thresholds applied to a similarity matrix between EEG channels. Each feature is recorded as a (birth, death) pair in a persistence diagram. To remain consistent with the hardware-efficient design, the similarity matrix is built from lightweight per-window summaries of each channel (e.g., windowed Pearson correlation) rather than from the full continuous signal, and the resulting persistence diagram is reduced to three scalar summaries — total persistence $P(t)$, maximum persistence, and bar count — rather than retained in full. This scalar reduction is our own hardware-efficiency choice; [9] retains the full diagram. The total persistence:

$$P = \sum_{(b_i, d_i) \in \text{Diag}} (d_i - b_i)$$

(3.6) Total persistence.

quantifies how topologically rich the network is. Pre-ictal collapse manifests as a decrease in P , indicating that the network's structural diversity is flattening into a single uniform cluster.

Collapse Operator (Ψ): The unified diagnostic metric that integrates the three axes above into a single weighted score:

$$\overline{MLZC}(t) = \frac{\mu_{MLZC} - MLZC(t)}{\sigma_{MLZC}} \quad \overline{PLV}(t) = \frac{PLV(t) - \mu_{PLV}}{\sigma_{PLV}} \quad \overline{P}(t) = \frac{\mu_P - P(t)}{\sigma_P}$$

(3.7a) Baseline normalization of the three axes.

$$\Psi(t) = w_1 \overline{MLZC}(t) + w_2 \overline{PLV}(t) + w_3 \overline{P}(t)$$

(3.7b) Collapse Operator.

Each axis enters as a standardized deviation from that patient’s inter-ictal baseline, but the three are oriented differently so that every term is positive in the direction the Complexity Collapse hypothesis predicts: complexity is written as a fall below baseline, coherence as a rise above it, and persistence as a fall below it (Eq. 3.7a). $\Psi(t)$ therefore rises as collapse approaches, and all three weights w_1, w_2, w_3 are expected to be positive. The baselines (μ, σ) are estimated per patient from inter-ictal windows in the training folds only, never from the held-out subject.

We adopt [6]’s three-axis form but use signed deviations where [6] uses absolute ones, since direction is the substantive content of the hypothesis. For why [6]’s weights cannot be assumed optimal here — imported coefficients, an approximate axis-to-role mapping, and an unvalidated baseline — see Section 4; Sections 6 and 7 evaluate them against re-tuned weights.

Soft Fusion: A post-processing layer that operates on the time series of operator scores $\Psi(t)$ [1]. Rather than triggering an alarm whenever a single window crosses a threshold, Soft Fusion maintains a rolling buffer of the most recent W window scores and computes the average of the top K values within that buffer:

$$S(t) = \frac{1}{K} \sum_{i \in \text{Top-}K(W)} \Psi_i$$

(3.8) Soft-Fusion score.

An alarm is issued only when $S(t)$ exceeds a stability threshold τ_{alarm} , suppressing isolated noise spikes while amplifying sustained pre-ictal trends [1]. Because this trades recall for precision, we expose it as a switchable layer: a default high-precision mode as described, and a high-sensitivity mode (in the limit $K = W = 1$) engaged manually or by an activity signal for high-risk contexts like swimming, where a missed seizure outweighs a false alarm. The switch reparametrizes only post-processing, leaving Ψ untouched; Stage 1 vs Stage 2 (Section 7) is compared in high-precision mode only.

3.3 Evaluation Methodology

Patient-wise (leave-one-subject-out) cross-validation: Train on $N - 1$ subjects, test on the held-out subject, repeat for each subject. This is the strictest generalization test for clinical EEG systems since the test subject’s idiosyncratic patterns are unseen during training. Per Section 2.4, chb01 and chb21 are held out together.

Event-Based F1 score: Standard F1 adapted for seizure prediction [11]: predictions are aggregated into events rather than scored per window. A True Positive is a predicted region overlapping an annotated seizure within an acceptable prediction horizon; a False Positive overlaps no seizure; a False Negative is an annotated seizure with no overlapping prediction. Event-based F1—the harmonic mean of event-level precision and recall—is the primary metric in this project. Per-window accuracy is uninformative due to the data imbalance noted in Section 2.4.

False Prediction Rate per hour (FPR/h): Average false alarms per hour of inter-ictal recording [1,11]. For wearable deployment, FPR/h must remain very low (typically below 0.15/h) to avoid alarm fatigue.

Seizure Prediction Horizon (SPH) and Seizure Occurrence Period (SOP): The two parameters that jointly define a valid prediction [1]. SPH is the minimum lead time between alarm and seizure onset (typically 1–5 minutes). SOP is the window after the SPH within which the seizure must occur for the alarm to count as a true positive (typically 5–30 minutes); a prediction outside the SOP is a false alarm. Given the committed $T_{\text{pre}} = 10$ minutes (Section 3.1), we set $\text{SPH} = 3$ min rather than the 1 min floor because three minutes suffices for the protective actions targeted here — leaving water, stopping a vehicle, moving from heights — while $\text{SOP} = 7$ min absorbs pre-ictal onset jitter; the split is a free parameter ($\text{SPH} + \text{SOP} = T_{\text{pre}}$) that Phase B will sweep.

3.4 Computational Cost

Table 3.2 below summarizes the structural simplifications introduced on each axis of the pipeline and the per-window saving each one produces, for the setting $N = 1,280$ samples, $C = 23$ channels. The reductions are of two kinds — per-window operation counts (MLZC, PLV) and retained-representation or memory size (persistent homology, per-window memory) — and are derived order-of-magnitude estimates, justified in the footnotes to the table; exact wall-clock and energy figures depend on the deployment platform and are a Phase B deliverable.

Table 3.2 — Sources of computational saving in the binarized pipeline.

Axis	Continuous-domain baseline	Our implementation	Reduction
MLZC	Real-valued Lempel-Ziv parse over continuous EEG	Integer/bitwise parse over binarized streams; no multiplication	$\sim 8\times$ compute
PLV	Complex-exponential evaluation ($\sin + \cos$), complex accumulation per sample, per channel	Bitwise comparison of phase bits for each channel pair	$\sim 100\times$ compute
Persistent homology	Full persistence diagram retained per window	Three scalar summary statistics retained per window	$\sim 30\times$ retained size
Per-window memory	Continuous EEG + analytic signal + persistence diagram (~ 350 kB)	Two one-bit-per-sample streams + three scalars (~ 7.4 kB)	$\sim 47\times$ memory

Both columns describe this project's own three axes (MLZC, PLV, persistent-homology summaries): the baseline is a full-precision implementation, included to quantify what binarization saves. Neither column describes how [9] or [6] computed their features.

MLZC ($\sim 8\times$). The LZ76 parse is $O(N)$ in both pipelines and its structure is unchanged; the saving is a constant factor. The continuous-domain baseline performs floating-point coarse-graining (the continuous-signal analogue of Eq. 3.4) and floating-point comparisons during the parse, whereas the binarized pipeline coarse-grains by integer summation (Eq. 3.4) and parses

integer sequences with no floating-point multiply. The $\sim 8\times$ figure is the typical cost ratio of a software floating-point operation to an integer/bit operation on the FPU-less embedded hardware targeted in Section 1.1.

PLV ($\sim 100\times$). For each of the $C(C-1)/2 = 253$ channel pairs the continuous-domain PLV evaluates $\exp(i(\phi_a[n] - \phi_b[n]))$ once per sample — two transcendental evaluations (sin, cos) plus a complex accumulation — while the binarized form reduces each channel's phase to one bit (Eq. 3.2) and accumulates pairwise bit agreement with a single bitwise operation per sample. The pair count and window length N are identical in both versions and cancel, so the saving is set by the per-sample operation ratio: on the order of 100 integer-operation-equivalents versus one bit operation. On hardware with a floating-point unit it is smaller.

Persistent homology ($\sim 30\times$). A reduction in retained representation, not in per-window operation count. A full persistence diagram for the window holds on the order of 45 finite birth–death pairs (≈ 90 float values in the configuration we expect); replacing it with three scalar summaries — total persistence $P(t)$ (Table 3.1), maximum persistence, and bar count — shrinks the stored topological feature by $\approx 30\times$. The persistence computation is additionally cheaper because it runs on a 23×23 inter-channel similarity matrix rather than the full window, but that factor is library- and filtration-dependent and is a Phase B deliverable.

Per-window memory ($\sim 47\times$). With $N = 1,280$ and $C = 23$ the window holds $N\cdot C = 29,440$ samples. Stored as float32 this is ≈ 115 kB of raw EEG plus ≈ 230 kB for the complex analytic signal plus a few kB for the persistence diagram, ≈ 350 kB in total. Each binary stream holds one bit per sample (≈ 3.7 kB), and Algorithm 2 retains two — SURE for the amplitude-binarized informational axis, Hilbert-phase for the coherence axis — so the binarized window is ≈ 7.4 kB plus three float scalars, giving $\approx 47\times$. A single retained stream would give $\approx 90\times$, but it could not supply both axes. The reduction remains one-to-two orders of magnitude.

4. Literature Review

Two of the features this project computes come from the hand-crafted family — the Phase-Locking Value [8] and Multiscale Lempel–Ziv Complexity [5,14] — and the principle it builds on comes from the complexity-framework family (Section 1.2). Across all three, no single body of work has demonstrated a system that is simultaneously interpretable, hardware-efficient, and validated under leave-one-subject-out evaluation with event-based metrics. That is the gap this project targets, and the studies below bear on it directly. Shoeb (2009) [10] — Introduced the CHB-MIT database in his MIT doctoral dissertation and, in the same work, presented the first detection system trained on it: a patient-specific Support Vector Machine on spectral energy features. Shoeb's SVM is the reference baseline against which essentially every later CHB-MIT system is compared. Two properties of that work propagate through the field: it is a detector rather than a predictor (it identifies ongoing ictal activity, not pre-ictal change), and its patient-specific protocol does not generalize to unseen subjects — a constraint our leave-one-subject-out validation explicitly stresses (Section 2.4).

- **Sirakov, Verma & Yang (2021) [11]** — Comprehensive survey of automated seizure detection and prediction systems evaluated on CHB-MIT, and the single most useful reference for mapping the field on this dataset. We rely on it for broad claims about CNN/RNN performance, evaluation protocols, and metric coverage on CHB-MIT. Two of its findings shape our pipeline directly: per-window sensitivities reported in the literature are inflated by patient-specific and per-window evaluation, and clinical metrics (FPR/h, SPH/SOP) are reported by only a minority of systems. Our verification plan (Section 7) reports these clinical metrics; our validation protocol (Section 3.3) avoids the inflation.
- **Balamas (2025) [1]** — Evaluates CNN backbones on CHB-MIT under a defined SPH and SOP, taking Short-Time Fourier Transform spectrograms as input. The central methodological contribution of the paper is a post-processing layer called Soft Fusion, which maintains a rolling buffer of recent pre-ictal probability scores and triggers an alarm only when the average of the top-K scores in that buffer exceeds a patient-tuned threshold. Soft Fusion is reported to improve sensitivity and alarm stability over a conventional k-of-n hard-voting rule. The deep backbone falls on the wrong side of the hardware gap of Section 1.3 and is not adopted here; the Soft Fusion layer, however, is decoupled from the upstream CNN and applied directly to our Collapse Operator score $\Psi(t)$ (Section 3.2, Eq. 3.8), preserving the alarm-stabilization benefit while collapsing upstream cost to three deterministic feature streams.
- **Myers et al. (2016) [8]** — Propose a seizure prediction and detection algorithm built around the Phase-Locking Value (PLV), computed between pairs of EEG electrodes local and remote to the suspected epileptic focus, with a companion Amplitude-Lock Value confirming ictal onset. The PLV serves as the pre-ictal marker: a sustained rise above a patient-specific synchrony threshold flags abnormal neuronal synchronization. The system is verified on CHB-MIT and reported to separate pre-ictal from ictal events. We borrow the empirical claim that PLV is a viable pre-ictal marker but differ in two respects: PLV is one of three axes integrated by the Collapse Operator rather than the sole signal, and we compute it from a phase-binarized stream (Eq. 3.5) rather than the full analytic signal of every channel pair, which removes most of the floating-point cost of the original formulation.
- **Yagis & Schiff (2016) [14]** — An early published evaluation of Multiscale Lempel–Ziv Complexity on CHB-MIT, motivated by the observation that classical LZC collapses on signals dominated by low-frequency rhythms, masking the high-frequency complexity changes that often precede seizure onset; coarse-graining the signal before the LZ parse exposes those changes. Despite a small cohort, the study

demonstrates that a single MLZC threshold separates ictal from non-ictal windows where single-scale LZC does not. Our use of MLZC as the informational axis of the Collapse Operator extends that observation in two directions: from detection to prediction, and from a small cohort to the full 22-subject CHB-MIT cohort under leave-one-subject-out validation.

- **Salazar Chon (2025) [9] and Kadhim [6]** — Salazar Chon's preprint introduces the Complexity Collapse hypothesis: a seizure is not neural chaos but a collapse of brain complexity into pathological order. He establishes this offline, on a few CHB-MIT records and a healthy control, using a three-metric "neurotopological signature" — permutation entropy for signal complexity, persistent homology over a Pearson-correlation connectivity network, and a discrete Ricci curvature for network geometry — separating healthy, inter-ictal, and ictal states with Kolmogorov–Smirnov tests and the bottleneck distance. Two properties matter here. It is an offline characterization of already-labelled states, not a predictor; its own future-work section names seizure prediction as the natural next step, which is precisely the problem this project takes up. And it defines no unified operator: the three metrics are compared statistically rather than fused into a single score, and it uses neither phase-locking coherence nor Lempel–Ziv complexity. Kadhim [6] contributes the operator form this project builds on — a weighted three-axis collapse operator, $\Delta\Phi = 0.40 \cdot |\Delta S| + 0.35 \cdot |\Delta I| + 0.25 \cdot |\Delta C|$, with baseline-deviation scoring and threshold bands. Its published code computes the three axes with the methods its documentation names: the structural axis combines Ollivier–Ricci curvature (via optimal-transport on the correlation graph) with H_1 persistent homology (via Ripser), the informational axis is permutation entropy, and the coherence axis is built from correlation-structure summaries. Three properties limit it as a basis for prediction, however. Its weights (0.40/0.35/0.25) are imported unchanged from an unrelated framework (FBD neurodynamic theory) rather than fit to seizure data. Its patient and healthy-control features are produced by inconsistent pipelines — controls substitute cheap proxies (a second-difference for curvature, sample entropy for homology) under the same feature names — so the baseline comparison is not like-for-like. And its baseline is global and its thresholds fixed, with no cross-validation. We therefore treat [6] as an unvalidated offline proof-of-concept.

5. Tools and Technologies

The implementation belongs to Phase B; this section commits to the tools it will use — a standard Python scientific-computing stack, chosen because every component has a mature, well-documented open-source library.

Table 5.1 — Tools and technologies used in the project.

Category	Tool
Language	Python
Numerical computing	NumPy, SciPy
EEG I/O	MNE-Python, pyEDFlib
Topology	Ripser / GUDHI
Machine learning	scikit-learn
Version control & collaboration	Git, GitHub
Plotting & analysis	Matplotlib, pandas
Development environment	Google colab

6. Flow Diagram

Figure 6.1 below expands the signal flow introduced in Figure 3.1 by adding the training-time procedure that selects between the two weight sets. The arrows show data flow from the CHB-MIT recordings through preprocessing and binarization, into the three feature axes (MLZC, PLV, and persistent homology), and finally through the two-stage weight comparison that selects the winning weight set. Algorithms 1 and 2 that follow expand the first two blocks of this diagram into explicit pseudocode; the feature-axis blocks are specified mathematically in Section 3.2 and the two-stage comparison is verified in Section 7.

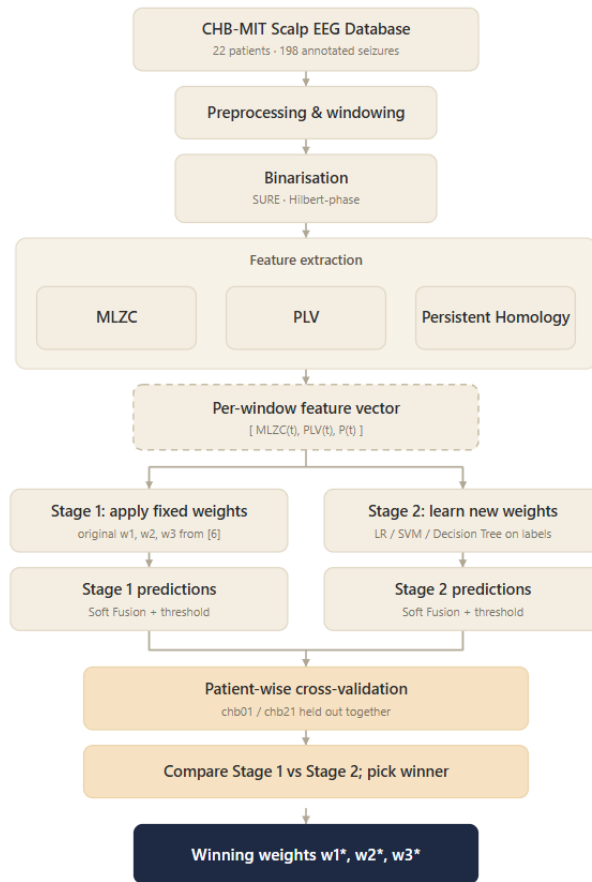


Figure 6.1

End-to-end pipeline of the proposed system, from raw CHB-MIT recordings through binarization, the three feature axes, the two-stage weight comparison, and the final selection of the winning weight set.

6.1 Pseudo-Code

The two algorithms below expand the 2nd and 3rd blocks of Figure 6.1 into explicit pseudocode. Algorithm 1 corresponds to the Preprocessing & windowing block: it walks the cohort patient by patient, restricts each patient to the EEG-only channel subset common to all of their EDF files (the non-uniform-channel issue documented for CHB-MIT in [3] and discussed in Section 2.4), and slides a window of length N at stride Δ within each file. Because the window loop is nested inside the file loop, no window crosses a file boundary — the inter-file-gap constraint also documented in [3] and discussed in Section 2.4 is enforced structurally rather than by an explicit check. Algorithm 2 corresponds to the Binarization block and is invoked once per window produced by Algorithm 1, returning the two binary representations — SURE thresholding [2,12] and Hilbert-phase extraction [4] — on which the three feature axes of Section 3.2 operate. The labelling function `assign_label` on line 16 of Algorithm 1 assigns each window to one of ictal, pre-ictal, or inter-ictal using the seizure annotations (provided with the CHB-MIT release [10]) and the pre-ictal horizon T_{pre} , and drops windows that straddle a state transition.

A window is treated as ambiguous, and therefore dropped, when it overlaps the seizure-onset boundary itself (so it is neither fully pre-ictal nor fully ictal) or the seizure-termination boundary (neither fully ictal nor fully inter-ictal). Dropping these windows prevents the classifier from being trained on signals whose true class is genuinely uncertain.

Algorithm 1 – Preprocessing & Windowing

```

1 Input: cohort of patients, valid EEG channel list  $E$ ,
2   window length  $N$ , stride  $\Delta$ , pre-ictal horizon  $T_{\text{pre}}$ 
3 Output: set of labelled windows  $\mathcal{W}$ 

4  $\mathcal{W} \leftarrow \emptyset$ 
5 for each patient  $s$  in cohort do
6    $C_s \leftarrow$  EEG channels common to all files of  $s$  (intersect with  $E$ )
7    $A_s \leftarrow$  seizure annotations of  $s$ 

8   for each EDF file  $f$  of patient  $s$  do
9      $X_f \leftarrow$  read file  $f$ , keeping only channels  $C_s$ 

10    for each window of length  $N$  at stride  $\Delta$  inside  $f$  do
11      label  $\leftarrow$  assign_label(window,  $A_s$ ,  $T_{\text{pre}}$ )
12      if label is not "ambiguous" then
13        add (window, label, patient_id) to  $\mathcal{W}$ 
14      end if
15    end for
16  end for
17 end for
18 return  $\mathcal{W}$ 

```

Figure 6.2

Algorithm 1: per-patient preprocessing and windowing of the CHB-MIT recordings. Expands the Preprocessing & windowing block of Figure 6.1.

Algorithm 2 – Binarization of one window

```
1 Input: window X with C channels of length N
2 Output: two binary windows B_SURE, B_Hilbert

3 for each channel c in X do
4   B_SURE[c] ← SURE_binarize(X[c])    (Eq. 3.1)
5   B_Hilbert[c] ← Hilbert_binarize(X[c]) (Eq. 3.2)
6 end for
7 return (B_SURE, B_Hilbert)
```

Figure 6.3

Algorithm 2: per-window binarization of the multi-channel EEG signal into SURE and Hilbert-phase binary streams. Expands the Binarization block of Figure 6.1; invoked once per window produced by Algorithm 1.

7. Verification Plan

This section describes how we will verify that the pipeline works correctly and produces clinically meaningful results. Verification is organized as a sequence of ten checkpoints (1–10) that follow the pipeline in order, summarized in Table 7.1. Each checkpoint describes what we are checking and what success looks like at that stage

Table 7.1 — Verification checkpoints, in pipeline order.

#	Step	What is verified	Success Criteria
1	Data ingestion	EEG recordings load correctly, non-brain channels are excluded, and gaps between recording files are not treated as continuous signal.	Every recording opens without errors, only valid EEG channels are kept, and short signal windows never cross a file boundary.
2	Annotation alignment	Each short window of EEG is correctly labelled as pre-ictal, ictal, or inter-ictal based on the clinical annotations.	Spot-checking a sample of labelled seizures against the original annotations shows the labels match the recorded timestamps.
3	Binarization	The two binarization methods (SURE and Hilbert-phase) produce binary streams that are balanced and varied, not stuck at a constant value.	On normal (inter-ictal) EEG, both methods produce binary streams with a healthy mix of zeros and ones, not all-zero or all-one outputs.
4	Feature axes	The three feature axes — complexity (MLZC), coherence (PLV), and topology (persistent homology) — produce sensible values that differ between normal and pre-seizure brain states.	Plots of each feature show visibly different distributions in inter-ictal versus pre-ictal windows for most patients; all values are finite and well-bounded.

5	Stage 1 (weights from [6])	The Collapse Operator using the weights of [6] behaves as the Complexity Collapse principle predicts — elevated near seizure onset, calm during normal activity.	For most patients, the operator score rises above its baseline ahead of seizures and returns to baseline between them.
6	Stage 2 (learned weights)	A transparent classifier (Logistic Regression, Linear SVM, or Decision Tree) can be successfully trained to combine the three features into a prediction score.	Training completes successfully and the learned weights are consistent with the Collapse Hypothesis — the model agrees that complexity drops and synchronization rises before seizures.
7	Cross-validation integrity	When one patient is held out for testing, none of that patient’s data appears in training — including the special case of chb01 and chb21, which are the same person recorded twice.	No training data ever appears in the held-out fold, and chb01 and chb21 are always held out together.
8	Soft Fusion	The smoothing layer correctly filters out brief noise spikes while preserving sustained pre-ictal trends.	An isolated spike alone does not trigger an alarm; only a sustained rise across several windows does.
9	Clinical metrics	The three clinical metrics — Event-Based F1, false alarms per hour, and prediction lead time — are calculated correctly per patient and across the whole cohort.	Metric calculations match hand-computed values on a small test case, and results are reported per-patient as well as cohort-averaged.
10	Stage comparison	Stage 1 (weights from [6]) and Stage 2 (learned weights) are compared fairly on the same held-out data, and the better-performing version is chosen.	A single winning weight set is reported, supported by the per-patient results, and the choice is reproducible from the same data.

7.1 Defining a meaningful result

Before stating the success conditions, we commit to the pre-ictal horizon $T_{\text{pre}} = 10$ minutes, which fixes the labelling rule of Algorithm 1 and the SPH/SOP split of Section 3.3. Ten minutes was chosen because it matches the evaluation horizon of Balamas [1] — the source of our Soft Fusion layer — and because the literature reports that pre-ictal complexity changes in CHB-MIT manifest most reliably within roughly the last 10 minutes before onset [8,14]. Longer horizons (e.g. 30 min) increase available lead time but introduce labels whose

pre-ictal character is contested in the literature; shorter horizons (e.g. 5 min) reduce class-imbalance pressure but compress SPH below the 1–5 min clinical band of Section 1.1.

The verification table above checks that the pipeline runs without bugs. But running without bugs is not the same as producing a scientifically meaningful result. For the project to count as a success, three things also need to hold.

- First, the Complexity Collapse principle has to survive our streaming realization: Stage 1 applies the weights of [6] to our own lightweight binarized features. If the operator score still rises before seizures and stays flat otherwise — for most patients — the binarization kept enough of the signal. If it doesn't, it was too lossy. Either outcome is informative, but only the first lets us claim a working lightweight operator.
- Second, whichever stage wins has to be useful: Because seizures are rare, a system that never alarms scores high accuracy by default, so beating "no alarm ever" is the minimum bar; the winner must also beat random alarms fired at our own rate. And if Stage 2 beats Stage 1 only marginally, we report both as a tie — declaring a winner by less than cross-validation noise would be dishonest.
- Third, the result has to be honest about who it works for: A 70% cohort average produced by working on 5 patients and failing on 17 is not a predictor; we therefore report per-patient results alongside the average. If Stage 2 wins, we also check its learned weights make biological sense: complexity should drop before seizures, synchronization should rise, topology should flatten. If the model learned the opposite, it is fitting an artifact and we cannot trust it.

If all three conditions hold — regardless of which stage wins — that's a success.

8. AI Tools Usages

Three AI tools were used during the preparation of this project, each for a distinct role, with explicit cross-checking between tools so that no single system's output was taken at face value.

Perplexity: Used early in the project to map the territory: what the Complexity Collapse hypothesis is, how it has been validated, and which datasets it has been tested on. Representative prompt: "What is the Complexity Collapse hypothesis in epileptic seizures, and which papers introduced it?"

Gemini: The Deep Research mode was used to expand the initial reading list into the full reference base, producing literature summaries and pointing to specific papers, definitions, and methods. Representative prompt: "Research Lempel–Ziv complexity for EEG seizure analysis — original definition, multiscale extension, and use as a pre-ictal biomarker."

Claude: Used to draft and restructure sections of the book against the agreed outline, and to independently re-check citations and quantitative claims that earlier tools had surfaced before those claims were accepted into the book. Representative prompt: "Audit Section 3 for citation gaps: flag every factual claim that does not have a citation, and propose which reference would be appropriate."

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